Title: IBMQueryDocs
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Date: 04-30-2026

At the beginning of the course, I honestly struggled a bit to decide what project to choose and how to use an AI approach in it. When we were asked to pick a project and explain what AI methods we would use, I felt really confused and not confident at all. Compared to a normal full-stack project where things are more straightforward, here I wasn't even sure how to start or what approach to follow. So I spent a lot of time just thinking and exploring options, and finally I decided to go with something simple so I could actually understand how everything works. That's when I chose a question-answering system with citations using the RAG approach. What I am most proud of is not just building the application, but how I started understanding and evaluating it deeply. After implementing the system, I didn't stop there, I started analyzing how well it was performing, like whether the retrieved chunks were correctly matching the relevant sources and how well the responses were being generated by the model. While doing this evaluation, I realized something important. I had reached a point where I could actually think critically about the system I built. The fact that I was able to design evaluation ideas like checking retrieval quality and model response quality made me feel that I truly understood the application. Compared to where I started, going from being unsure about even choosing an AI approach to being able to evaluate my own system confidently is what I'm most proud of.

During this project, I first tried to understand how a RAG application works and what all steps are involved. After that, I divided the work into weekly tasks and followed a simple plan to complete it step by step. I also kept some buffer time at the end for testing, because I was new to this AI framework and expected some last-minute issues. The in-class presentations were very helpful because everyone shared their progress and we could ask questions. This helped me learn new things from others and clear my doubts during development. For tools and approaches, I primarily used Jupyter Notebook for development and experimentation, Git for version tracking, and structured my work in a task-based manner similar to agile sprint planning. This helped me stay organized and track what was completed versus what still needed attention. At the end, I faced an issue when converting the Jupyter Notebook into Python modules while running it in Google Colab. I couldn't properly load and run the model. I discussed it with a classmate, and I learned that they were using an inference API instead of Google Colab, which runs the model directly on Hugging Face. This helped me understand a better approach. If I do this project again, I will start talking to others earlier when working on something new, so I can understand different approaches and avoid last-minute issues.

During this project, I faced a few challenges, but I kept learning step by step using GenAI tools, YouTube, documentation, and help from classmates. At the beginning, I was not very clear about what a RAG application is. So, I used simple videos and GenAI explanations to understand it better. Slowly I understood retrieval, embeddings, and generation, and that helped me start building the system. One major issue I faced was that even after building the full RAG pipeline, the LLM was only giving very short answers (around 3–4 words) and also taking a lot of time. I was confused whether it was because of retrieval or because of the model. So I tested it by removing the retrieval part and directly giving the full PDF to the model with a question. Even then I got the same type of short output. That is when I understood the problem was not with chunks, but with the model. One important thing I learned was not to blindly trust GenAI. When it suggested an embedding model, I asked why it is used and checked Lang Chain documentation. I also kept asking things like "why this model?" and "is there a better option?" and compared answers. Earlier I knew model size matters, but I didn't know how much

difference it makes in real use. When I switched from Flan-T5 to Mistral 7B, I clearly saw better and more meaningful outputs.

During the in-class presentation, I explained my contribution to the RAG project in a simple way. I talked about how I built the retrieval part using basic similarity search and then improved it using MMR so the chunks are more relevant and not repeated. I also explained how I connected retrieval with the LLM to generate answers. I shared my testing experience as well. I noticed that Flan-T5 was giving very short answers, so I tried Mistral-7B-Instruct. I didn't just change it randomly, I tested both and saw the difference clearly in output quality. I also mentioned the IBM WatsonxDocsQA dataset and why I used it to evaluate the system. What I am most proud of is that I could actually show how each change improved the results. I didn't just explain the project, I also explained what I tried, what failed, and what worked. I was able to show everything clearly using a table to represent improvement in a numeric way, which made it easy for everyone to understand. I also presented my code and explained my testing logic, so people could see exactly how I verified the results. And I was able to present it in a way that everyone could easily understand.

Self-Assessment:

1. Gain technical expertise and strengthen AI system development knowledge and skills you've learned in the course – **100%**
2. Engage in effective project management activities and communication, facilitated by team meetings and human interactions and supported by appropriate use of project management tools -**100%**
3. Demonstrate responsibility for your own learning in your use of external resources, including generative AI -**100%**
4. Communicate effectively and knowledgeably your individual contribution to the project. – **100%**
5. Reflect on your project learning to justify how you have achieved each of the learning goals. – **100%**